

AR SnowAssist – Project Report (Phase 1)

1. Topic Selection

Selected Topic: Augmented Reality (AR) Based Application for Social Benefit. Project Title: AR SnowAssist – Real-Time Snow Removal & Accessibility Navigation System. This project proposes the development of a full-stack augmented reality application that assists pedestrians, elderly individuals, and people with mobility challenges during winter conditions. Users can report snow-blocked sidewalks by uploading images and location data. The system stores reports and provides AR-based overlays to guide users toward safer walking paths.

2. Problem Statement

During winter seasons, snow accumulation blocks sidewalks and pedestrian pathways, creating accessibility and safety challenges. Existing municipal systems primarily track snowplow vehicles rather than pedestrian-level accessibility. There is a lack of real-time, user-generated reporting systems that visually guide pedestrians using augmented reality.

3. System Overview

- User Registration and Authentication System
- Snow Report Submission (Image + GPS Coordinates)
- Database Storage and Report Management
- Web Dashboard for Viewing Reports
- AR Mode for Real-Time Visual Navigation Assistance

4. Domain Model

The domain model consists of three primary entities: User: - userId - name - email - password - role - createdAt SnowReport: - reportId - userId (Foreign Key) - imageURL - latitude - longitude - description - status (Pending / Verified / Resolved) - createdAt Relationship: One User can create multiple SnowReports. Each SnowReport belongs to exactly one User.

5. High-Level System Architecture

Frontend: - HTML, CSS, JavaScript (React in future phase) - WebAR interface using Three.js or A-Frame Backend: - Node.js - Express.js - MongoDB - Multer for image uploads Data Flow: User -> Frontend -> Backend API -> Database Backend -> AR Engine -> Visual Overlay on Camera

6. Product Vision Statement

For pedestrians and individuals with mobility challenges Who need safe navigation during winter conditions, The AR SnowAssist system Is a full-stack augmented reality navigation application That provides real-time snow condition updates and AR-based safe route guidance, Unlike existing municipal snow tracking systems, Our product focuses on pedestrian accessibility, user-generated reporting, and real-time AR visualization.